MA104 · DISCRETE MATH · LECTURE 3

Mathematical Induction

$P(k) \Rightarrow P(k+1)$

Lecture notes and seven worked proofs — sums, telescoping series, divisibility

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1 · The idea behind induction

Mathematical induction is a method of proof. It is used to prove statements that claim something is true **for every positive integer** n — formulas for sums, divisibility facts, inequalities, properties of algorithms. Checking $n = 1, 2, 3, \ldots$ one by one can never finish; induction replaces infinitely many checks with **two** finite arguments.

✎ LECTURE NOTE · THE DOMINO PICTURE

Imagine an endless row of dominoes, one for each positive integer. To be sure that **all** of them fall, you need exactly two facts:

1. the **first** domino falls, and
2. the dominoes are spaced so that **whenever** one falls, it knocks over the next.

Neither fact alone is enough. Fact 2 without fact 1 describes a perfectly arranged row that nobody pushes; fact 1 without fact 2 knocks over a single domino. Together they force domino 1, then 2, then 3, and so on forever. In induction, "domino n falls" is the statement $P(n)$.

Formally, let $P(n)$ be a statement about the positive integer n. The **principle of mathematical induction** says:

$$\underbrace{P(1)}_{\text{basis step}} \quad \wedge \quad \underbrace{\forall k \geq 1 \, (P(k) \to P(k+1))}_{\text{inductive step}} \quad \Longrightarrow \quad \forall n \geq 1 \, P(n)$$

Key idea. In the inductive step we do **not** prove $P(k)$. We prove the **implication** "if $P(k)$ is true, then $P(k+1)$ is true". That is why we are allowed to **assume** $P(k)$ — the assumption is called the **inductive hypothesis** (IH).

2 · The three-step template

Every proof in this lecture follows the same pattern:

Step	What you write	What it means
1	Show the statement is true for $n = 1$.	The first domino falls.
2	Assume it is true for $n = k$.	Suppose domino k falls (the IH).
3	Prove it is true for $n = k+1$, **using** step 2.	Domino k knocks over domino $k+1$.

✎ LECTURE NOTE · HOW TO ATTACK STEP 3 FOR A SUM

For an identity of the form $a_1 + a_2 + \ldots + a_n = f(n)$ the inductive step is almost mechanical:

1. **Write the goal first.** Substitute $k+1$ for n in the formula, so you know exactly which expression $f(k+1)$ you are aiming for.
2. **Add the next term.** Add a_{k+1} to both sides of the IH. The left side becomes the sum up to $k+1$; the right side becomes $f(k) + a_{k+1}$.
3. **Simplify toward the goal.** Factor out what $f(k+1)$ has in common with $f(k)$ — usually a factor like $(k+1)$ or $\frac{1}{k+1}$ — rather than expanding everything.
4. **Conclude.** Say explicitly that the result equals $f(k+1)$, so $P(k+1)$ holds.

Note. Throughout these notes, LHS and RHS mean the left-hand side and right-hand side of the formula. In the basis step compute them **separately** and then compare — never start by writing "LHS = RHS", since that is what you are trying to show.

3 · Sums of integers and squares

Example 1 Sum of the first n integers

Prove by mathematical induction that, for all positive integers n,

$$1 + 2 + 3 + \ldots + n = \frac{1}{2}n(n+1)$$

STEP 1 · BASIS STEP

For $n = 1$: LHS $= 1$, RHS $= \frac{1}{2}(1)(2) = 1$. $\therefore$ LHS $=$ RHS $= 1$, so the formula holds for $n = 1$.

STEP 2 · INDUCTIVE HYPOTHESIS

Assume the formula is true for $n = k$:

$$1 + 2 + 3 + \ldots + k = \frac{1}{2}k(k+1)$$

STEP 3 · INDUCTIVE STEP

We must show $1 + 2 + \ldots + (k+1) = \frac{1}{2}(k+1)(k+2)$. Adding $(k+1)$ to both sides of the IH:

$$\begin{aligned} 1 + 2 + 3 + \ldots + k + (k+1) &= \frac{1}{2}k(k+1) + (k+1) \quad \text{(by the IH)} \\ &= (k+1)\left(\frac{k}{2} + 1\right) \qquad \text{factor out } (k+1) \\ &= \frac{1}{2}(k+1)(k+2). \end{aligned}$$

The right member $\frac{1}{2}(k+1)(k+2)$ is the original formula with $k+1$ substituted for n. Hence the formula is true for every positive integer n.

✎ LECTURE NOTE

Reading the conclusion. The handout spells out the domino chain: the formula is true for $n = 1$; since it is true for $n = 1$ it is true for the next integer $n = 2$; since it is true for $n = 2$ it is true for $n = 3$; and so on. Steps 1 and 3 together are exactly that chain.

A second proof (Gauss). Write the sum forwards and backwards and add: each of the n columns sums to $n + 1$, so twice the sum is $n(n + 1)$. This explains **where** the formula comes from; induction only **verifies** a formula you already have.

Sanity check (computed by Typst):

n	LHS (added up)	RHS (formula)	equal?
1	1	1	✓
2	3	3	✓
3	6	6	✓
4	10	10	✓
5	15	15	✓
6	21	21	✓

Example 2 Sum of the first n squares

Use mathematical induction to prove that, for all positive integers n,

$$1^2 + 2^2 + 3^2 + \ldots + n^2 = \frac{1}{6}n(n+1)(2n+1)$$

STEP 1 · BASIS STEP

For $n = 1$: $\text{LHS} = 1^2 = 1$, $\text{RHS} = \frac{1}{6}(1)(2)(3) = 1$. $\therefore \text{LHS} = \text{RHS} = 1$.

STEP 2 · INDUCTIVE HYPOTHESIS

Assume the formula is true for $n = k$:

$$1^2 + 2^2 + 3^2 + \ldots + k^2 = \frac{1}{6}k(k+1)(2k+1)$$

STEP 3 · INDUCTIVE STEP

Goal: $1^2 + \ldots + (k+1)^2 = \frac{1}{6}(k+1)(k+2)(2k+3)$. Adding $(k+1)^2$ to both sides of the IH:

$$\begin{aligned}
1^2 + 2^2 + \ldots + k^2 + (k+1)^2 &= \frac{1}{6}k(k+1)(2k+1) + (k+1)^2 \quad \text{(by the IH)} \\
&= \frac{1}{6}(k+1)[k(2k+1) + 6(k+1)] \text{ factor out } \frac{1}{6}(k+1) \\
&= \frac{1}{6}(k+1)[2k^2 + k + 6k + 6] \\
&= \frac{1}{6}(k+1)[2k^2 + 7k + 6] \\
&= \frac{1}{6}(k+1)(k+2)(2k+3).
\end{aligned}$$

The right member $\frac{1}{6}(k+1)(k+2)(2k+3)$ is the original formula with $k+1$ substituted for n. Therefore the formula is true for every positive integer n.

✎ LECTURE NOTE

Where does $(k+2)(2k+3)$ come from? We **know** the target, so we know the quadratic $2k^2+7k+6$ must factor as $(k+2)(2(k+1)+1)=(k+2)(2k+3)$. Multiply it out to check: $2k^2+3k+4k+6$ ✓. Writing the goal down first turns factoring into checking.

Why factor instead of expand? Expanding $\frac{1}{6}k(k+1)(2k+1)$ into a cubic and then adding $(k+1)^2$ works, but you must then re-factor a cubic. Pulling out the common factor $(k+1)$ immediately leaves only a quadratic.

n	LHS (added up)	RHS (formula)	equal?
1	1	1	✓
2	5	5	✓
3	14	14	✓
4	30	30	✓
5	55	55	✓
6	91	91	✓

4 · Telescoping sums

Example 3 A sum of reciprocals

Prove by mathematical induction:

$$\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\frac{1}{3\cdot 4}+\ldots+\frac{1}{n(n+1)}=\frac{n}{n+1},\quad n\geq 1$$

STEP 1 · BASIS STEP

For $n=1$: LHS $=\frac{1}{1\cdot 2}=\frac{1}{2}$, RHS $=\frac{1}{1+1}=\frac{1}{2}$. $\therefore$ LHS $=$ RHS $=\frac{1}{2}$.

STEP 2 · INDUCTIVE HYPOTHESIS

Assume the formula is true for $n=k$:

$$\frac{1}{1\cdot 2}+\frac{1}{2\cdot 3}+\ldots+\frac{1}{k(k+1)}=\frac{k}{k+1}$$

STEP 3 · INDUCTIVE STEP

Goal: the sum up to $\frac{1}{(k+1)(k+2)}$ equals $\frac{k+1}{k+2}$. Adding $\frac{1}{(k+1)(k+2)}$ to both sides of the IH:

$$\begin{aligned}
\frac{1}{1\cdot 2}+\ldots+\frac{1}{k(k+1)}+\frac{1}{(k+1)(k+2)} &= \frac{k}{k+1}+\frac{1}{(k+1)(k+2)} && \text{(by the IH)}\\
&= \frac{1}{k+1}\left(k+\frac{1}{k+2}\right) && \text{factor out } \frac{1}{k+1}\\
&= \frac{1}{k+1}\left(\frac{k^2+2k+1}{k+2}\right)\\
&= \frac{1}{k+1}\frac{(k+1)(k+1)}{k+2} && k^2+2k+1=(k+1)^2\\
&= \frac{k+1}{k+2}.
\end{aligned}$$

The right member $\frac{k+1}{k+2}$ is the original formula with $k+1$ substituted for n. Therefore the formula is true for every positive integer n.

✎ LECTURE NOTE · WHY IT TELESCOPES

Each term splits into a difference (partial fractions):

$$\frac{1}{i(i+1)} = \frac{1}{i} - \frac{1}{i+1}.$$

Adding them, almost everything cancels in pairs:

$$\left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \ldots + \left(\frac{1}{n} - \frac{1}{n+1}\right) = 1 - \frac{1}{n+1} = \frac{n}{n+1}.$$

This is a direct proof of the same identity and a good way to **discover** the formula. Notice also that the sum is always less than 1, however large n is.

n	LHS (added up)	RHS (formula)	equal?
1	$\frac{1}{2}$	$\frac{1}{2}$	✓
2	$\frac{2}{3}$	$\frac{2}{3}$	✓
3	$\frac{3}{4}$	$\frac{3}{4}$	✓
4	$\frac{4}{5}$	$\frac{4}{5}$	✓
5	$\frac{5}{6}$	$\frac{5}{6}$	✓
6	$\frac{6}{7}$	$\frac{6}{7}$	✓

Example 4 Reciprocals of odd products

Prove by mathematical induction:

$$\frac{1}{1\cdot 3} + \frac{1}{3\cdot 5} + \frac{1}{5\cdot 7} + \ldots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}, \quad n \geq 1$$

STEP 1 · BASIS STEP

For $n = 1$: LHS $= \frac{1}{1\cdot 3} = \frac{1}{3}$, RHS $= \frac{1}{2+1} = \frac{1}{3}$. $\therefore$ LHS $=$ RHS $= \frac{1}{3}$.

STEP 2 · INDUCTIVE HYPOTHESIS

Assume the formula is true for $n = k$:

$$\frac{1}{1\cdot 3} + \frac{1}{3\cdot 5} + \ldots + \frac{1}{(2k-1)(2k+1)} = \frac{k}{2k+1}$$

STEP 3 · INDUCTIVE STEP

The next term is obtained by putting $n = k+1$ in $\frac{1}{(2n-1)(2n+1)}$, which gives $\frac{1}{(2k+1)(2k+3)}$. Goal: the new sum equals $\frac{k+1}{2k+3}$. Adding $\frac{1}{(2k+1)(2k+3)}$ to both sides of the IH:

$$\begin{aligned}
\frac{1}{1\cdot 3}+\ldots+\frac{1}{(2k-1)(2k+1)}+\frac{1}{(2k+1)(2k+3)} &= \frac{k}{2k+1}+\frac{1}{(2k+1)(2k+3)} && \text{(by the IH)}\\
&= \frac{1}{2k+1}\left(k+\frac{1}{2k+3}\right) && \text{factor out } \frac{1}{2k+1}\\
&= \frac{1}{2k+1}\left(\frac{2k^2+3k+1}{2k+3}\right)\\
&= \frac{1}{2k+1}\,\frac{(2k+1)(k+1)}{2k+3} && \text{factor the quadratic}\\
&= \frac{k+1}{2k+3}.
\end{aligned}$$

The right member $\frac{k+1}{2k+3}$ is the original formula with $k+1$ substituted for n. Therefore the formula is true for every positive integer n.

✎ LECTURE NOTE

Finding the next term. The most frequent error here is writing the $(k+1)$-th term as $\frac{1}{(2k)(2k+2)}$ or $\frac{1}{(2k+1)(2k+2)}$. Always substitute carefully: $2(k+1)-1 = 2k+1$ and $2(k+1)+1 = 2k+3$.

Telescoping again. $\frac{1}{(2i-1)(2i+1)} = \frac{1}{2}\left(\frac{1}{2i-1}-\frac{1}{2i+1}\right)$, so the sum is $\frac{1}{2}\left(1-\frac{1}{2n+1}\right) = \frac{n}{2n+1}$ — same pattern as Example 3.

n	LHS (added up)	RHS (formula)	equal?
1	$\frac{1}{3}$	$\frac{1}{3}$	✓
2	$\frac{2}{5}$	$\frac{2}{5}$	✓
3	$\frac{3}{7}$	$\frac{3}{7}$	✓
4	$\frac{4}{9}$	$\frac{4}{9}$	✓
5	$\frac{5}{11}$	$\frac{5}{11}$	✓
6	$\frac{6}{13}$	$\frac{6}{13}$	✓

5 · A geometric sum

Example 5 A geometric sum

Use mathematical induction to prove that

$$2^1+2^2+2^3+\ldots+2^n = 2(2^n-1), \quad n \geq 1$$

STEP 1 · BASIS STEP

For $n=1$: LHS $= 2$, RHS $= 2(2-1) = 2$. $\therefore$ LHS $=$ RHS $= 2$.

STEP 2 · INDUCTIVE HYPOTHESIS

Assume the formula is true for $n = k$:

$$2^1+2^2+2^3+\ldots+2^k = 2\left(2^k-1\right)$$

STEP 3 · INDUCTIVE STEP

Goal: $2^1 + ... + 2^{k+1} = 2\big(2^{k+1} - 1\big)$. Adding 2^{k+1} to both sides of the IH:

$$\begin{aligned} 2^1 + 2^2 + ... + 2^k + 2^{k+1} &= 2\big(2^k - 1\big) + 2^{k+1} \quad \text{(by the IH)} \\ &= 2^{k+1} - 2 + 2^{k+1} \quad 2 \cdot 2^k = 2^{k+1} \\ &= 2 \cdot 2^{k+1} - 2 \\ &= 2\big(2^{k+1} - 1\big). \end{aligned}$$

The right member $2\big(2^{k+1} - 1\big)$ is the original formula with $k+1$ substituted for n. Therefore the formula is true for every positive integer n.

✎ LECTURE NOTE

This is a special case of the geometric series

$$a + ar + ar^2 + ... + ar^{n-1} = a\frac{r^n - 1}{r - 1}, \quad r \neq 1,$$

with $a = 2$, $r = 2$. In computing terms: $2^1 + ... + 2^n = 2^{n+1} - 2$, which is why an n-bit register of all 1s holds $2^n - 1$.

The only algebra fact used in step 3 is $2 \cdot 2^k = 2^{k+1}$ — exponent rules are the usual tool when the sum has powers.

n	LHS (added up)	RHS (formula)	equal?
1	2	2	✓
2	6	6	✓
3	14	14	✓
4	30	30	✓
5	62	62	✓
6	126	126	✓

6 · Divisibility

✎ LECTURE NOTE · THE DIVISIBILITY STRATEGY

"d divides m" (written $d \mid m$) means $m = dq$ for some integer q. For a statement "$d \mid f(n)$ for all $n \geq 1$" the inductive step has a different shape from the sums:

1. Expand $f(k+1)$.
2. **Rearrange** it so that $f(k)$ appears inside it: $f(k+1) = f(k) + (...)$ or $f(k+1) = c \cdot f(k) + (...)$.
3. $f(k)$ is a multiple of d by the IH; show the leftover "(...)" is visibly a multiple of d too.
4. A sum of multiples of d is a multiple of d, so $d \mid f(k+1)$.

Example 6 $n^3 + 2n$ is divisible by 3

Prove that for any positive integer n, $n^3 + 2n$ is divisible by 3.

STEP 1 · BASIS STEP

For $n = 1$: $1^3 + 2(1) = 1 + 2 = 3$, and 3 is divisible by 3.

STEP 2 · INDUCTIVE HYPOTHESIS

Assume the statement is true for $n = k$: $k^3 + 2k$ is divisible by 3.

STEP 3 · INDUCTIVE STEP

Prove that $(k+1)^3 + 2(k+1)$ is divisible by 3:

$$\begin{aligned}(k+1)^3 + 2(k+1) &= k^3 + 3k^2 + 3k + 1 + 2k + 2 \\ &= k^3 + 2k + 3k^2 + 3k + 3 \qquad \text{regroup} \\ &= (k^3 + 2k) + 3(k^2 + k + 1).\end{aligned}$$

Each part is divisible by 3:

- $k^3 + 2k$ is divisible by 3 (by the IH)
- $3(k^2 + k + 1)$ is divisible by 3 (it is 3 times an integer).

Hence $(k+1)^3 + 2(k+1)$ is also divisible by 3.

By induction, $3 \mid n^3 + 2n$ for every positive integer n.

✎ LECTURE NOTE

The regrouping is the whole proof. After expanding, we deliberately **look for** $k^3 + 2k$ (the IH) inside the expression, collect it, and check that what is left has a factor of 3.

Without induction: $n^3 + 2n = n^3 - n + 3n = (n-1)n(n+1) + 3n$. The product of three consecutive integers is always divisible by 3, and so is $3n$.

n	$n^3 + 2n$	factored	multiple of 3?
1	3	3×1	✓
2	12	3×4	✓
3	33	3×11	✓
4	72	3×24	✓
5	135	3×45	✓
6	228	3×76	✓

Example 7 $3^n - 1$ is a multiple of 2

Prove that for any positive integer n, $3^n - 1$ is a multiple of 2.

STEP 1 · BASIS STEP

For $n = 1$: $3^1 - 1 = 3 - 1 = 2$, and 2 is a multiple of 2.

STEP 2 · INDUCTIVE HYPOTHESIS

Assume the statement is true for $n = k$: $3^k - 1$ is a multiple of 2.

STEP 3 · INDUCTIVE STEP

Prove that $3^{k+1}-1$ is a multiple of 2:

$$\begin{aligned} 3^{k+1}-1 &= 3\cdot 3^k - 1 \\ &= 2\cdot 3^k + 1\cdot 3^k - 1 \text{ split } 3 = 2+1 \\ &= 2\cdot 3^k + \left(3^k - 1\right). \end{aligned}$$

Each part is a multiple of 2:

- $2\cdot 3^k$ is a multiple of 2 (it is 2 times an integer);
- 3^k-1 is a multiple of 2 (by the IH).

Hence $3^{k+1}-1$ is also a multiple of 2.

By induction, 3^n-1 is a multiple of 2 for every positive integer n.

Correction to the handout. The original handout ends step 3 with "hence $3^{k+1}-1$ is also divisible by 3". It should read **divisible by 2** — indeed $3^{k+1}-1$ is never divisible by 3. (In Example 6 the handout also repeats the line "Step 1" twice; it is printed once here.)

✎ LECTURE NOTE

The splitting trick. Writing $3 = 2+1$ produces the IH expression 3^k-1 plus a term with an obvious factor 2. The same trick proves the general fact $(a-1) \mid (a^n-1)$: write $a\cdot a^k - 1 = (a-1)a^k + \left(a^k - 1\right)$.

Without induction: 3^n is a product of odd numbers, so it is odd, and an odd number minus 1 is even.

n	3^n-1	factored	multiple of 2?
1	2	2×1	✓
2	8	2×4	✓
3	26	2×13	✓
4	80	2×40	✓
5	242	2×121	✓
6	728	2×364	✓

7 · Pitfalls to avoid

Common mistake. Skipping the basis step. The step $P(k)\to P(k+1)$ can hold for a false statement. Example: "$n+1=n$". If $k+1=k$, then adding 1 gives $k+2=k+1$, so the inductive step "works" — but $P(1)$ $(2=1)$ is false, and so is every $P(n)$.

Common mistake. Assuming what you want to prove. Step 2 assumes $P(k)$ for **one** arbitrary k, not "for all n". Writing "assume the formula is true for all n" in step 2 makes the proof circular.

Common mistake. Not using the hypothesis. If step 3 never uses the IH, it is either a direct proof (fine, but then it is not induction) or, more often, it is wrong. Mark the place where the IH is used, as with (by the IH) in these notes.

Common mistake. **Working on both sides at once.** Do not start step 3 from the equation you want to prove and manipulate both sides until you get $0 = 0$. Start from one side (the sum up to $k+1$) and transform it, line by line, into the other.

Common mistake. **Checking examples is not a proof.** The sanity-check tables in these notes make the formulas believable; they prove nothing. $n^2 + n + 41$ is prime for $n = 0, 1, ..., 39$, yet $40^2 + 40 + 41 = 41^2$ is not.

Rule. A complete induction proof has **three** labelled parts: the basis step (computed, not assumed), the inductive hypothesis (for a single k), and the inductive step, in which the hypothesis is visibly used to reach the statement for $k+1$.

8 · Practice

Prove each statement by mathematical induction for all integers $n \geq 1$.

1. $1 + 3 + 5 + ... + (2n-1) = n^2$

2. $1^3 + 2^3 + ... + n^3 = \left(\frac{1}{2}n(n+1)\right)^2$

3. $1 \cdot 2 + 2 \cdot 3 + ... + n(n+1) = \frac{1}{3}n(n+1)(n+2)$

4. $1 + 3 + 3^2 + ... + 3^{n-1} = \frac{1}{2}(3^n - 1)$

5. $\frac{1}{2 \cdot 3} + \frac{1}{3 \cdot 4} + ... + \frac{1}{(n+1)(n+2)} = \frac{n}{2(n+2)}$

6. $1 \cdot 1! + 2 \cdot 2! + ... + n \cdot n! = (n+1)! - 1$

7. $6 \mid n^3 - n$

8. $4 \mid 5^n - 1$

9. $3 \mid 2^{2n} - 1$

10. $2^n > n$

Note. **Hints.** (5) The next term is $\frac{1}{(k+2)(k+3)}$; factor out $\frac{1}{k+2}$. (7) Expand $(k+1)^3 - (k+1)$ and look for $k^3 - k$; the leftover is $3k(k+1)$, and $k(k+1)$ is even. (8) Split $5 = 4 + 1$ as in Example 7. (10) From $2^k > k$ get $2^{k+1} = 2 \cdot 2^k > 2k \geq k + 1$.